

## Trigonometric identities



1  $\frac{\sin x}{\cos x}$

2 0

3 a  $\frac{1}{2}$

b  $\frac{a}{c}$

4 a  $\frac{8}{17}$

b  $-\frac{12}{13}$

5 a  $\frac{4}{3}$

b  $\frac{2}{\sqrt{5}}$

6 a  $\sin \theta$

b  $\sin y$

c  $\tan \theta - 1$

7 a 2

b  $\frac{17}{31}$

c  $\frac{11}{6}$

d  $\pm \frac{1}{\sqrt{2}}$

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## Pythagorean identities

8 a Yes

b No

9

a 1



b 4

10

a  $-\sin^2(\theta)$

b  $1 - 2 \cos \theta \sin \theta$

c  $-\cos^3(\theta)$

d  $10 - 6 \cos x$

e  $\tan^2(\theta)$

f  $\cos^2(\theta)$

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$$\frac{x^2}{16} + \frac{y^2}{9} = 1$$



## Proofs

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|--------------------------------|--------------------------------|
| <b>a</b> No solution provided. | <b>b</b> No solution provided. |
| <b>c</b> No solution provided. | <b>d</b> No solution provided. |
| <b>e</b> No solution provided. | <b>f</b> No solution provided. |
| <b>g</b> No solution provided. | <b>h</b> No solution provided. |
| <b>i</b> No solution provided. | <b>j</b> No solution provided. |

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|--------------------------------|--------------------------------|
| <b>a</b> No solution provided. | <b>b</b> No solution provided. |
| <b>c</b> No solution provided. |                                |